# Project Proposal

**Problem Selection:**
The problem that I selected for this project is chemical classification using Raman spectroscopy data. The goal of this project is to correctly identify chemical compound from Raman spectrum. I am doing research on Pump-Probe spectroscopy and optical analysis so this project is somewhat related to my research topic.

**Dataset:**
I'll use an open-source Raman spectral dataset. It consists of 3510 samples from 32 chemical compounds which are commonly used in Active Pharmaceutical Ingredients. Each spectrum contains 3276 features. (from intensity vs Raman shift).
This dataset is sufficient for this project. While this dataset is not extremely large, but it is enough to train and validate a model.

**Deep Network:**
I'll use 1D-CNN for classification of Raman spectra into 32 classes. Raman spectrum measures intensity against a single variable, the Raman shift. For every specific shift ($cm^{-1}$) there is only one intensity value. CNN can learn important local features such as sharp peaks, peak widths, neighboring spectral patterns. The model will use a few convolutional layers to learn patterns from the data. It'll have with activation functions and pooling layers to process the information. After that, it will use one or more fully connected layers to make the final classification. I will start with a basic 1D-CNN model. If needed, I'll add simple improvements like dropout or batch normalization to help the model perform better and avoid overfitting.

**Framework:**
I'll use PyTorch to implement the model. It's not difficult to understand. Good for learning.

**Reference Materials:**
To build sufficient background I'll use Dr. Hagan's Neural Network Desing: Deep learning book. I'll also use Dr. Hagan's class lectures. I'll try to read research papers on Raman spectroscopy and spectral classification.

**Performance Evaluation:**
I'll try to evaluate the model using Accuracy, Confusion matrix and F1 score. The dataset will be split into training, validation and test sets.

**Project Schedule**
**Week 1:**
- Understand dataset
- Download data (CSV loading, normalization)

**Week 2:**
- Implement basic neural network in PyTorch
- Train and debug model

**Week 3 & 4:**

Tune hyperparameters (learning rate, hidden size)

Add improvements (normalization, dropout)

**Week 5:**

Evaluate results

Prepare report and plots